Reviewer Pack — Minimal Rank of Twisted Group Representations over Disjointn…

Entry 408a7082b621 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-26 19:52:38 UTC

Minimal Rank of Twisted Group Representations over Disjointness Communication Complexity

Entry ID: `408a7082b621` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-05-26 19:52:38 UTC

1. Conjecture

Field A (mathematical branch): Twisted Group Representation Theory **Field B** (complexity object): Communication Complexity: Disjointness

Statement:

> [‘For any disjointness communication protocol C, there exists a twisted group representation R such that the minimal rank of the induced matrix representation is at least Ω(n^2 log n), where n is the number of bits in the input.’, ‘For all disjointness protocols C, τ(R) ≥ Ω(n^2 log n)’, ‘Where τ(R) denotes the minimal rank of the twisted group representation induced by protocol C.’]

Rationale (proposer’s reasoning):

> [‘Twisted group representations have been used to study communication complexity in the context of entanglement and quantum information theory. This conjecture explores an extension of this concept into classical communication complexity, potentially revealing new structural insights.’, ‘The minimal rank of a representation is a fundamental invariant that captures the complexity of the underlying mathematical object. If this conjecture holds, it would establish a direct link between the algebraic structure of twisted group representations and the computational complexity of disjointness.’]

Taxonomy category: `COMM_DISJ` (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `2cafefc3d545949c`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> The conjecture is supported if, for all disjointness protocols C with n bits of input, the minimal rank τ(R) of the twisted group representation R induced by C satisfies τ(R) ≥ Ω(n^2 log n) AND the difference between τ(R) and n^2 log n is within a 5% margin. The conjecture is falsified if any protocol C has an induced τ(R) < Ω(n^2 log n) or if the difference exceeds a 5% margin.

3. Barrier filter (F1)

Final: `PASS`

Barrier	Verdict	Confidence	LLM$_1$	LLM$_2$
RELATIVIZATION	SAFE	1.00	UNCERTAIN	SAFE
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	0.80	HITS	UNCERTAIN
KARP_LIPTON	SAFE	1.00	SAFE	UNCERTAIN

4. Novelty audit

Verdict: `NOVEL` against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    n = random.randint(5, 40)
    protocol = [random.choice([0, 1]) for _ in range(n)]

    # Define a simple mapping from protocol to group elements
    group_elements = {i: (i % 2, i // 2) for i in range(n)}
    R = [[group_elements[protocol[i]] if protocol[i] == protocol[j] else (0, 0) for j in range(n)]

    # Calculate the minimal rank of the matrix representation
    def gaussian_elimination(matrix):
        m, n = len(matrix), len(matrix[0])
        for i in range(m):
            max_row = i
            for j in range(i+1, m):
                if abs(matrix[j][i]) > abs(matrix[max_row][i]):
                    max_row = j
            matrix[i], matrix[max_row] = matrix[max_row], matrix[i]
            if matrix[i][i] == 0:
                continue
            factor = 1 / matrix[i][i]
            for j in range(n):
                matrix[i][j] *= factor
            for j in range(m):
                if j != i:
                    factor = matrix[j][i]
                    for k in range(n):
                        matrix[j][k] -= factor * matrix[i][k]
        rank = sum(1 for row in matrix if any(row))
```

```
        return rank

    τ_R = gaussian_elimination(R)

    # Check the conjecture
    expected_rank = n**2 * math.log(n, 2)
    margin = expected_rank * 0.05
    conjecture_holds = τ_R >= expected_rank - margin and τ_R <= expected_rank + margin

    return {
        "metric_name": "minimal_rank",
        "metric_value": τ_R,
        "instances_tested": n,
        "conjecture_holds": conjecture_holds,
        "counterexample": f"rank={τ_R}, expected={expected_rank}" if not conjecture_holds else ""
    }

if __name__ == "__main__":
    seeds = [int(x) for x in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3
    results = []

    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value)**2 for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_frac
    elif any(not r["conjecture_holds"] for r in results) and support_fraction >= 0.8:
        first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"{r['counterexample']}\" first_failing_seed={fir
    else:
        print("RESULT: INCONCLUSIVE")
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e4cc353a.py", line 70, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e4cc353a.py", line 50, in run_trial
    τ_R = gaussian_elimination(R)
          ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e4cc353a.py", line 34, in gaussian_elim
    if abs(matrix[j][i]) > abs(matrix[max_row][i]):
       ^^^^^^^^^^^^^^^^^
```

```
TypeError: bad operand type for abs(): 'tuple'
```

8. Critic adversarial review

Critic verdict: `CHALLENGE`

Critic reasoning:

> skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: `INCONCLUSIVE`

Reasoning:

> The test crashed before producing data, which means that it was unable to verify the conjecture's conditions. | next: Re-run the test with a different seed or on a different system to ensure stability and obtain results.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	11317
2	preregistration	ollama_remote	glm4:latest	0	0	6684
3	novelty	ollama_remote	glm4:latest	0	0	4720
4	novelty	ollama_remote	glm4:latest	0	0	5457
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	23380
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10184
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11051
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9340
9	judge	ollama_remote	glm4:latest	0	0	10557

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 92690 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/408a7082b621.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/408a7082b621.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/408a7082b621.tar.gz` (if generated)